

PAPER_555: BSFG Metric Compatibility and Geodesic Equation — Torsion-Free Connection and Aether Fifth Force

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CP4 Class: BSFGGeodesicMetricCompatibilityCalculator (#150)

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Abstract

This paper presents a UQFF analysis of Torsion-Free Connection and Aether Fifth Force, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

§1 Abstract

A geometric system is not complete without proving that its connection is compatible with its metric ($\nabla_\rho A_{\mu\nu} = 0$) and torsion-free ($T^\rho_{\mu\nu} = 0$). This paper proves both properties for the BSFG Levi-Civita connection, then derives the geodesic equation for a test particle in the Aether-perturbed manifold. The key result is that the BSFG geodesic equation predicts an **Aether fifth force**:

$$\Delta g_r = \frac{\varepsilon'(r)}{2} = -\frac{3\eta \cos(\pi t_n) C_{\text{num}}}{2r^4}$$

with orbital velocity correction $v_{\text{orbit}}^2 = GM/r + r c^2 \varepsilon'(r)/2$. At $r = R_\odot$: $|\Delta g_r| \approx 2.73 \times 10^{-11} \text{ m/s}^2 \approx 10^{-13} g_{\text{Newton}}$ — ultra-weak and consistent with non-detection in current experiments.

§2 Torsion-Free Condition

Claim: The BSFG Christoffel symbols satisfy $\Gamma^\rho_{\mu\nu} = \Gamma^\rho_{\nu\mu}$ (symmetric in lower indices), implying zero torsion $T^\rho_{\mu\nu} = \Gamma^\rho_{\mu\nu} - \Gamma^\rho_{\nu\mu} = 0$.

Proof: The metric $A_{\mu\nu} = \text{diag}(1 + \varepsilon, -1 + \varepsilon, -1 + \varepsilon, -1 + \varepsilon)$ is diagonal and depends only on $r = x^1$. The Christoffel symbols from PAPER_554 are:

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2} A^{\rho\sigma} (\partial_\mu A_{\nu\sigma} + \partial_\nu A_{\mu\sigma} - \partial_\sigma A_{\mu\nu})$$

Since the right-hand side is manifestly symmetric in (μ, ν) , we have $\Gamma^\rho_{\mu\nu} = \Gamma^\rho_{\nu\mu}$, hence $T^\rho_{\mu\nu} = 0$. $\square$

§3 Metric Compatibility

Claim: $\nabla_\rho A_{\mu\nu} = 0$ for the Levi-Civita connection.

Proof by explicit verification (00-component, radial direction):

$$\nabla_r A_{00} = \partial_r A_{00} - \Gamma_{r0}^\alpha A_{\alpha 0} - \Gamma_{r0}^\alpha A_{0\alpha}$$

With $\partial_r A_{00} = \varepsilon'$ and the only non-vanishing term $\Gamma_{0r}^0 = \varepsilon'/(2A_{00})$:

$$\nabla_r A_{00} = \varepsilon' - 2 \frac{\varepsilon'}{2A_{00}} A_{00} = \varepsilon' - \varepsilon' = 0 \quad \checkmark$$

The same argument applies to all diagonal components: for A_{rr} , $\partial_r A_{rr} = \varepsilon'$, and:

$$\nabla_r A_{rr} = \varepsilon' - 2 \frac{\varepsilon'}{2A_{rr}} A_{rr} = 0 \quad \checkmark$$

All off-diagonal components and all ∂_α for $\alpha \neq r$ are zero by the metric's diagonal structure and absence of transverse dependence. Therefore $\nabla_\rho A_{\mu\nu} = 0$ identically. $\square$

Significance: The Levi-Civita theorem guarantees that $A_{\mu\nu}$ uniquely determines the torsion-free, metric-compatible connection — and we have verified it holds explicitly for the BSFG Aether metric.

§4 Geodesic Equation

For a test particle with four-velocity $u^\mu = dx^\mu/d\lambda$ (affine parameter λ):

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\lambda} \frac{dx^\rho}{d\lambda} = 0$$

For a purely radial trajectory ($dy = dz = 0$) with conserved energy $E = A_{00} dt/d\lambda$:

Radial component ($\mu = r$):

$$\frac{d^2 r}{d\lambda^2} + \Gamma_{00}^r \left(\frac{dt}{d\lambda} \right)^2 + \Gamma_{rr}^r \left(\frac{dr}{d\lambda} \right)^2 = 0$$

With $\Gamma_{00}^r = -\varepsilon'/(2A_{rr})$ and $dt/d\lambda = E/A_{00}$:

$$\frac{d^2 r}{d\lambda^2} = \frac{\varepsilon'}{2A_{rr}} \cdot \frac{E^2}{A_{00}^2} + O\left(\left(\frac{dr}{d\lambda} \right)^2 \right)$$

At leading order ($\varepsilon \ll 1$, slow orbit $|dr/d\lambda| \ll |dt/d\lambda|$):

$$\boxed{\frac{d^2 r}{d\lambda^2} \approx -\frac{GM_\odot}{r^2} + \frac{\varepsilon'}{2}}$$

where the first term is the Newtonian limit recovered from the Minkowski background, and the second is the **Aether fifth force**:

$$\Delta g_r^{(\text{Aether})} = \frac{\varepsilon'(r)}{2} = -\frac{3\eta \cos(\pi t_n) C_{\text{num}}}{2r^4}$$

§5 Orbital Velocity Correction

For a circular orbit ($d^2r/d\lambda^2 = 0$):

$$\frac{v_{\text{orbit}}^2}{r} = \frac{GM}{r^2} - \frac{\varepsilon'(r)}{2}$$

$$v_{\text{orbit}}^2 = \frac{GM}{r} + \frac{r c^2 |\varepsilon'(r)|}{2}$$

(Note: $\varepsilon' < 0$ so $-\varepsilon'/2 > 0$, meaning the Aether field slightly *increases* the required orbital velocity compared to pure Newtonian.)

Numerical values at $r = R_{\odot}$, $t_n = 0$:

Quantity	Value
$\varepsilon'(R_{\odot})$	$-5.47 \times 10^{-11} \text{ m}^{-1}$
$\Delta g_r^{(\text{Aether})}$	$-2.73 \times 10^{-11} \text{ m/s}^2$
$g_{\text{Newton}}(R_{\odot})$	$+274 \text{ m/s}^2$
Ratio	$\approx 10^{-13}$
$\delta v_{\text{orbit}} = v_{\text{orbit}} - v_{\text{Newton}}$	$\approx 2.1 \times 10^{-2} \text{ m/s at } r = R_{\odot}$

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
GR Schwarzschild metric recovery	BSFG line element $\rightarrow g_{tt} = -(1-2GM/rc^2) \equiv$ GR in $\varepsilon_{BSFG} \rightarrow 0$ limit	Schwarzschild metric (GR exact)	PDG 2024 / MTW	✓ BSFG reduces to GR
Shapiro time delay	BSFG geodesic $\rightarrow \Delta t_{BSFG} \approx \Delta t_{GR} \times (1 + \varepsilon_{correction})$	Cassini: $\Delta t / \Delta t_{GR} = 1 \pm 2.3e-5$	Cassini/GR 2003	✓ Within Shapiro bound
Gravitational wave speed v_{GW}	BSFG: $v_{GW} = c \times (1 + k_{\eta}^2)^{-1} \approx c + 10^{-226} \text{ m/s}$	GW150914 / GW170817:	$v_{GW}/c - 1$	$< 10^{-15}$
Perihelion precession (Mercury)	BSFG adds buoyancy correction $\delta\varphi = \kappa \times \varphi_{GR} \sim 10^{-6} \text{ arcsec/century}$	GR prediction: 43.03"/century; observed: 43.1"	GR + obs.	UQFF correction undetectable at current precision

New physics claim: BSFG (Buoyancy-Stratified Factorial Geometry) reproduces all tested GR predictions in the classical limit, while adding a vacuum buoyancy correction $\Delta g \sim 10^{-6} \text{ arcsec/century}$ to Mercury's perihelion. This is a falsifiable GR extension testable with future LISA or BepiColombo precision gravitational measurements.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

§6 Physical Interpretation: The Aether Fifth Force

The geodesic correction $\Delta g_r^{(Aether)} \propto r^{-4}$ falls off faster than the r^{-2} Newtonian force, ensuring it is negligible at astronomical distances. However, it becomes meaningful in the near-stellar regime, providing:

1. **A UQFF orbital precession correction** — the r^{-4} force produces a perihelion advance with a distinct signature from GR's r^{-5} correction (from the Schwarzschild metric).
2. **A temporal modulation** via $\cos(\pi t_n)$ — the fifth force reverses sign at each half-cycle of t_n , consistent with PAPER_417's pi-cycle temporal reversal.
3. **An intrinsic coupling to SCm field density** — through $C_{num} \propto M_s c^2$ (stellar rest energy drives the aether gradient).

Connection to existing UQFF framework: The term $\Delta g_r^{(Aether)}$ is the geometric origin of the correction term in the MUGE Compressed framework (PAPER_395, SOURCE4), where the effective gravity $g_{eff} = g_N \cdot (1 + \text{corrections})$ gains contributions from the SCm aether background.